

## SECTION A

$$1 \text{ a) } f(1) = \lim_{x \rightarrow 1^-} x+1 = \lim_{x \rightarrow 1^+} \frac{x^2 - x - a}{x - a} \quad (B1)$$

$$\frac{1^2 - 1 - a}{1 - a} = 2 = \frac{1^2 - 1 - a}{1 - a}$$

$$\frac{-a}{1-a} = 2 \quad (K1)$$

$$-a = 2 - 2a$$

$$a = 2 \quad (J1)$$

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x+1)(x-2)}{x-2} \quad (K1) \checkmark$$

$$= \lim_{x \rightarrow 2} x+1 \quad \left. \begin{array}{l} (K1) \\ (J1) \end{array} \right\}$$

$$= 2+1$$

$$= 3$$

$$b) \lim_{x \rightarrow +\infty} \left( \frac{2x+5}{3x-4} \right)^{\frac{1}{3}} = \lim_{x \rightarrow +\infty} \left( \frac{\frac{2x+5}{x}}{\frac{3x-4}{x}} \right)^{\frac{1}{3}} \quad (K1)$$

$$= \lim_{x \rightarrow +\infty} \left( \frac{2 + \frac{5}{x}}{3 - \frac{4}{x}} \right)^{\frac{1}{3}}$$

$$= \left( \frac{2+0}{3-0} \right)^{\frac{1}{3}} \quad (K1)$$

$$= \left( \frac{2}{3} \right)^{\frac{1}{3}} \text{ or } \sqrt[3]{\frac{2}{3}} \quad (J1)$$

2 a)  $x = t^{-1} + 3t^2$ ,  $y = t^2 - 4t^{\frac{1}{2}}$ ,  $t > 0$   
 $\frac{dx}{dt} = -t^{-2} + 6t$   $\frac{dy}{dt} = 2t - 2t^{-\frac{1}{2}}$

(KI) Find both  
 Either 1 correct

$$= -\frac{1}{t^2} + 6t$$

$$= 2t - \frac{2}{t^{1/2}}$$

$$= \frac{-1 + 6t^3}{t^2}$$

$$= \frac{2t^{3/2} - 2}{t^{1/2}}$$

$$\frac{dy}{dx} = \frac{2t - \frac{2}{t^{1/2}}}{-\frac{1}{t^2} + 6t}$$

or

$$\frac{dy}{dx} = \frac{2t^{3/2} - 2}{t^{1/2}} \cdot \frac{t^2}{-1 + 6t^3}$$

$$= \frac{2t^3 - 2t^{3/2}}{-1 + 6t^3}$$

(JI)

$$\frac{d^2y}{dx^2} = \frac{(-1 + 6t^3)(6t^{1/2} - 3t^{-1/2}) - (2t^3 - 2t^{3/2})(18t^2)}{(-1 + 6t^3)^2} \cdot \frac{t^2}{-1 + 6t^3}$$

(KI)  
 (JI)

$$\left. \frac{d^2y}{dx^2} \right|_{t=1} = \frac{5(3) - (0)}{5^2} \cdot \frac{1}{5}$$

(KI) ✓

$$= \frac{3}{25} \text{ or } 0.12$$

(JI)

b)  $y = \ln(\sin 2x) - \ln(1 - \cos 2x)$

$$\frac{dy}{dx} = \frac{2 \cos 2x}{\sin 2x} - \frac{2 \sin 2x}{1 - \cos 2x}$$

(KI) atas (BOTH)

(KI) bwh (BOTH)

(JI)

$$= \frac{2 \cos 2x - 2 \cos^2 2x - 2 \sin^2 2x}{(\sin 2x)(1 - \cos 2x)}$$

$$= \frac{2 \cos 2x - 2(\cos^2 2x + \sin^2 2x)}{(\sin 2x)(1 - \cos 2x)}$$

(KI)

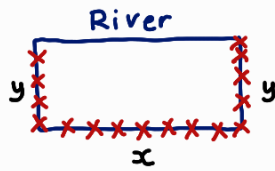
$$= \frac{-2(1 - \cancel{\cos 2x})}{(\sin 2x)(1 - \cancel{\cos 2x})}$$

$$= \frac{-2}{\sin 2x}$$

$$= -2 \operatorname{cosec} 2x$$

(JI)

3)



$$x + 2y = 100 \quad \text{(B1)}$$

Maximum Area

$$\begin{aligned} A &= x y \\ &= (100 - 2y) y \\ &= 100y - 2y^2 \end{aligned}$$

$$\frac{dA}{dy} = 100 - 4y = 0 \quad \text{(K1) } \checkmark \text{ diff correctly}$$

$$y = 25 \quad \text{(J1)}$$

$$\frac{d^2A}{dy^2} = -4 < 0 \quad (\text{Maximum}) \quad \text{(K1) } \checkmark$$

$$\begin{aligned} \therefore y &= 25 \\ x &= 100 - 2(25) = 50 \end{aligned} \quad \left. \vphantom{\begin{aligned} \therefore y &= 25 \\ x &= 100 - 2(25) = 50 \end{aligned}} \right\} \text{(J1)}$$

## SECTION B

$$\begin{aligned} 1 \text{ a) } \ln 2^{4x-1} &= \ln 8^{x+5} + \log_2 16^{1-2x} \\ &= \ln 2^{3(x+5)} + \log_2 2^{4(1-2x)} \end{aligned} \quad \textcircled{K1} \textcircled{J1}$$

$$(4x-1) \ln 2 = (3x+15) \ln 2 + 4-8x$$

$$4x \ln 2 - \ln 2 = 3x \ln 2 + 15 \ln 2 + 4 - 8x$$

$$x(\ln 2 + 8) = 4 + 16 \ln 2$$

$$x = \frac{4 + 16 \ln 2}{8 + \ln 2}$$

Ⓚ1 attempt to solve  
x = 8 ada ln 2

Ⓜ1

$$\text{b) } z_3 = \frac{2+3i}{4+2i} \cdot \frac{4-2i}{4-2i}$$

Ⓚ1 KETAT

$$= \frac{8 - 4i + 12i - 6i^2}{16 - 4i^2}$$

Ⓚ1

$$= \frac{14}{20} + \frac{8}{20}i$$

$$= \frac{7}{10} + \frac{2}{5}i$$

Ⓜ1

$$|z_3| = \sqrt{\left(\frac{7}{10}\right)^2 + \left(\frac{2}{5}\right)^2}$$

Ⓚ1

$$= \frac{\sqrt{65}}{10}$$

Ⓜ1

$$2 \text{ a) } \frac{4x}{4-x} \geq 6$$

$$\frac{4x}{4-x} - 6 \geq 0 \quad \text{(KI)}$$

$$\frac{4x - 24 + 6x}{4-x} \geq 0 \quad \text{(KI)}$$

$$\frac{10x - 24}{4-x} \geq 0 \quad \text{(JI)}$$

|                   | $\frac{12}{5}$            | 4                   |               |
|-------------------|---------------------------|---------------------|---------------|
| Interval of $x$   | $(-\infty, \frac{12}{5}]$ | $[\frac{12}{5}, 4)$ | $(4, \infty)$ |
| Test value of $x$ | 0                         | 3                   | 5             |
| $10x - 24$        | -                         | +                   | +             |
| $4 - x$           | +                         | +                   | -             |
| Conclusion        | -                         | $\oplus$            | -             |

(KI) • any method  
• ada +-

$$\text{Solution set : } \left\{ x : \frac{12}{5} \leq x < 4 \right\}$$

(JI)

$$b) |x-6| < 3|x-2|$$

$$\left( \sqrt{(x-6)^2} \right)^2 < \left( 3\sqrt{(x-2)^2} \right)^2$$

$$(x-6)^2 < 9(x-2)^2 \quad \text{(KI) SBS}$$

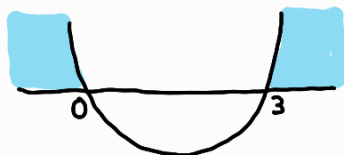
$$x^2 - 12x + 36 < 9(x^2 - 4x + 4) \quad \text{(KI) expand BOTH CORRECTLY}$$

$$8x^2 - 24x > 0$$

$$8x(x-3) > 0$$

$$x(x-3) > 0$$

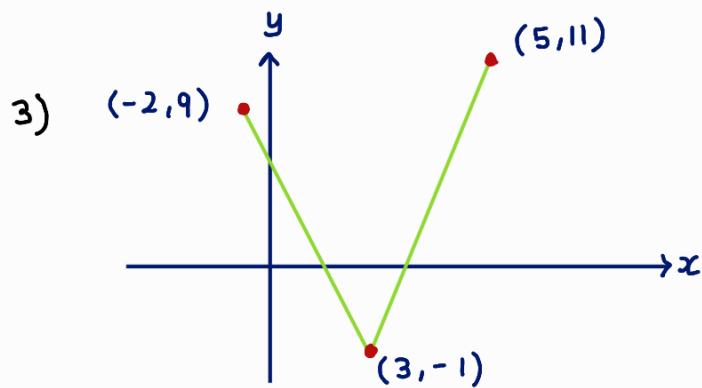
← (JI) either



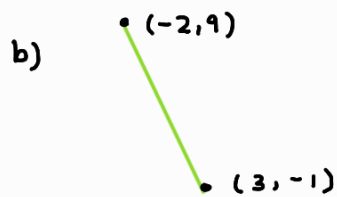
(KI) ✓

(KI) ✓ correct region

$$\text{Solution set : } \left\{ x : x < 0 \text{ OR } x > 3 \right\} \quad \text{(JI)}$$



a)  $R_f = [-1, 11]$  (B1)

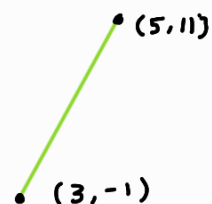


$$m_1 = -2$$

$$y + 1 = -2(x - 3) \quad \text{or}$$

$$y - 9 = -2(x + 2)$$

$$y = -2x + 5$$



$$m_2 = 6$$

$$y + 1 = 6(x - 3)$$

$$y - 11 = 6(x - 5)$$

$$y = 6x - 19$$

(K1) • find BOTH  
• either 1 ✓

(K1) • find BOTH  
• either 1 ✓

(J1) BOTH

$$f(x) = \begin{cases} -2x + 5, & -2 \leq x \leq 3 \\ 6x - 19, & 3 \leq x \leq 5 \end{cases}$$

(J1) • bentuk piecewise  
• accept  $\leq < \leq$   
 $< \leq \leq$   
• tak terima  $<$   
 $<$

c)  $f \circ f(0) = f(5)$   
 $= 6(5) - 19$   
 $= 11$  } (K1)  
 (J1)

$$4 \text{ a) } f \circ g(x) = \frac{3}{x-5}, \quad x \neq 5$$

$$\frac{3}{[g(x)]^2 - 4} = \frac{3}{x-5}$$

$$[g(x)]^2 - 4 = x - 5$$

$$[g(x)]^2 = x - 1$$

$$g(x) = \pm \sqrt{x-1}$$

$$g(x) = \sqrt{x-1}, \quad g(x) \geq 0$$

$$D_g = [1, \infty)$$

(KI)

(KI) attempt to find  $g(x)$

(JI)

(BI)

$$b) \quad g(x_1) = g(x_2)$$

$$\sqrt{x_1 - 1} = \sqrt{x_2 - 1}$$

$$x_1 = x_2$$

Since  $x_1 = x_2$ ,  $g^{-1}(x)$  exists.

}

(KI)

(JI)

$$g \circ g^{-1}(x) = x$$

$$\sqrt{g^{-1}(x) - 1} = x$$

$$g^{-1}(x) = x^2 + 1$$

$$D_{g^{-1}} = R_g = [0, \infty)$$

(KI)

(JI)

(BI)

$$c) \quad h \circ g^{-1}(x) = 5x^2 + 12$$

$$h(x^2 + 1) = 5x^2 + 12$$

$$\text{Let } u = x^2 + 1$$

$$x^2 = u - 1$$

$$\therefore h(u) = 5(u-1) + 12 \quad (\text{KI})$$

$$= 5u + 7 \quad (\text{JI})$$

$$h(x) = 5x + 7 = ax + b \quad (\text{KI})$$

$$\therefore a = 5, \quad b = 7 \quad (\text{JI})$$

Method 2

$$h \circ g^{-1}(x) = a(x^2 + 1) + b \quad (\text{KI})$$

$$= ax^2 + a + b \quad (\text{JI})$$

$$ax^2 + a + b = 5x^2 + 12 \quad (\text{KI})$$

$$\therefore a = 5 \quad (\text{JI})$$

$$b = 7$$

}

$$5 \text{ a) } f(x) = \frac{2x^2+2}{x^2-4}$$

$$x^2 - 4 = 0$$

$$x = -2, x = 2 \quad (\text{B1})$$

$$\lim_{x \rightarrow -2^-} \frac{2x^2+2}{x^2-4} = +\infty \quad \left. \vphantom{\lim_{x \rightarrow -2^-} \frac{2x^2+2}{x^2-4}} \right\} (\text{K1})$$

$$\lim_{x \rightarrow -2^+} \frac{2x^2+2}{x^2-4} = -\infty$$

$$\lim_{x \rightarrow 2^-} \frac{2x^2+2}{x^2-4} = -\infty \quad \left. \vphantom{\lim_{x \rightarrow 2^-} \frac{2x^2+2}{x^2-4}} \right\} (\text{K1})$$

$$\lim_{x \rightarrow 2^+} \frac{2x^2+2}{x^2-4} = +\infty$$

$\therefore$  Vertical asymptotes are  $x = -2$  and  $x = 2$ . (J1) (J1)

$$\lim_{x \rightarrow +\infty} \frac{\frac{2x^2+2}{x^2}}{x^2-4} \quad \left. \vphantom{\lim_{x \rightarrow +\infty} \frac{\frac{2x^2+2}{x^2}}{x^2-4}} \right\} (\text{K1}) \lim_{x \rightarrow \infty} \quad (\text{K1}) \div x^2$$

$$= \lim_{x \rightarrow +\infty} \frac{2 + \frac{2}{x^2}}{1 - \frac{4}{x^2}}$$

$$= \frac{2+0}{1-0}$$

$$= 2$$

$$\lim_{x \rightarrow -\infty} \frac{\frac{2x^2+2}{x^2}}{x^2-4}$$

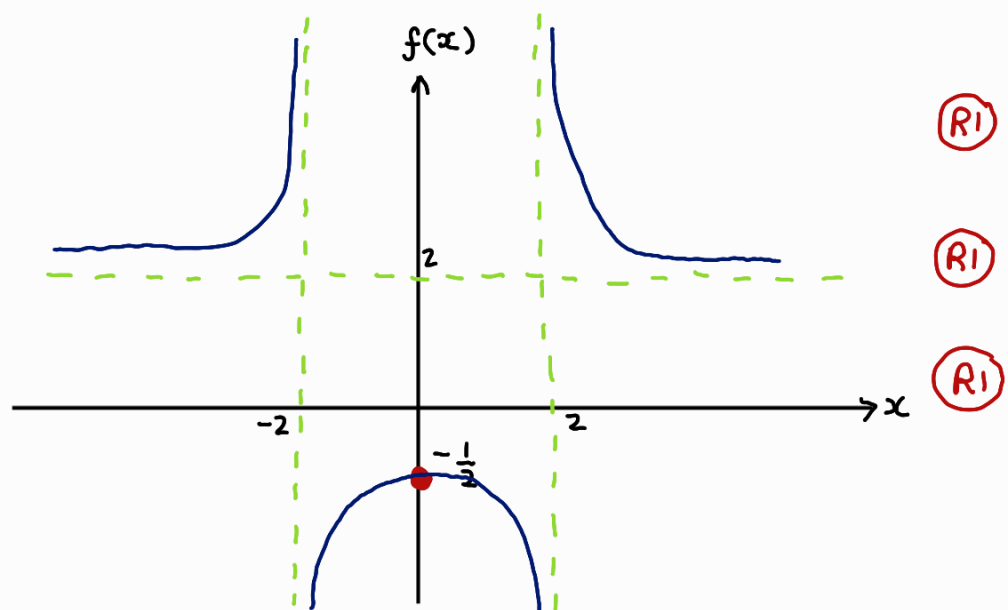
$$= \lim_{x \rightarrow -\infty} \frac{2 + \frac{2}{x^2}}{1 - \frac{4}{x^2}}$$

$$= \frac{2+0}{1-0}$$

$$= 2$$

$\therefore$  Horizontal asymptote is  $y = 2$  (J1)





$$6 \text{ a) } y = \tan\left(\frac{\pi}{4} + x\right)$$

$$\frac{dy}{dx} = \sec^2\left(\frac{\pi}{4} + x\right) \quad \text{(KI)}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 2 \left[ \sec\left(\frac{\pi}{4} + x\right) \right]' \cdot \frac{d}{dx} \sec\left(\frac{\pi}{4} + x\right) \\ &= 2 \sec\left(\frac{\pi}{4} + x\right) \sec\left(\frac{\pi}{4} + x\right) \tan\left(\frac{\pi}{4} + x\right) \quad \text{(KI) (JI)} \\ &= 2 \underbrace{\sec^2\left(\frac{\pi}{4} + x\right)}_{\frac{dy}{dx}} \underbrace{\tan\left(\frac{\pi}{4} + x\right)}_y \end{aligned}$$

$$\frac{d^2y}{dx^2} = 2y \frac{dy}{dx} \quad \text{(KI)}$$

$$\therefore \frac{d^2y}{dx^2} - 2y \frac{dy}{dx} = 0 \quad \text{(JI)}$$

$$\begin{aligned} \text{b) } g'(x) &= \frac{(e^x - 2)(e^x) - (e^x)(e^x)}{(e^x - 2)^2} \quad \text{(KI) (JI)} \\ &= \frac{e^x(e^x - 2 - e^x)}{(e^x - 2)^2} \\ &= -\frac{2e^x}{(e^x - 2)^2} \end{aligned}$$

$$g'(x) = -2$$

$$\frac{-2e^x}{(e^x - 2)^2} = -2$$

$$\text{Let } e^x = y$$

$$-2y = -2(y - 2)^2$$

$$-2y = -2(y^2 - 4y + 4)$$

$$2y^2 - 10y + 8 = 0$$

$$y^2 - 5y + 4 = 0$$

$$(y - 1)(y - 4) = 0$$

$$\therefore y = 1 \quad \text{or} \quad y = 4$$

$$e^x = 1$$

$$x = 0$$

$$e^x = 4$$

$$x = \ln 4$$

(K1) ✓ • attempt (darab silang & expand)

(J1)

} (K1) ✓

(J1)

$$c) \quad y^2 = x^3$$

$$2y \frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{2y}$$

$$\left. \frac{dy}{dx} \right|_{(1, -1)} = -\frac{3}{2}$$

$$\left. \frac{dy}{dx} \right|_{(1, 1)} = \frac{3}{2}$$

$$2x^2 + 3y^2 = k$$

$$4x + 6y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{3y}$$

$$\left. \frac{dy}{dx} \right|_{(1, -1)} = \frac{2}{3}$$

$$\left. \frac{dy}{dx} \right|_{(1, 1)} = -\frac{2}{3}$$

(K1) • diff betul  
• ganti at least  
1 point

(J1)

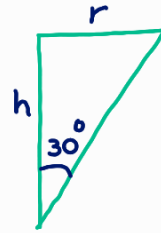
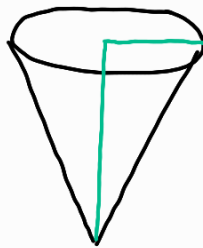
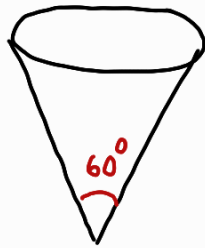
(K1)

(J1)

Relationship : perpendicular / reciprocal / normal / orthogonal

(J1)

7)



$$\rightarrow \frac{dV}{dt} = -5$$

$$\rightarrow \tan 30^\circ = \frac{r}{h}$$

$$\frac{\sqrt{3}}{3} = \frac{r}{h} \quad \text{(B1)}$$

$$\rightarrow V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left( \frac{\sqrt{3}}{3} h \right)^2 h \quad \text{(K1) ✓}$$

V dlm sebutan h

$$V = \frac{\pi h^3}{9}$$

$$\frac{dV}{dh} = \frac{\pi h^2}{3}$$

(K1) ✓  
diff correctly

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$-5 = \frac{\pi h^2}{3} \cdot \frac{dh}{dt} \quad \text{(K1) ✓}$$

$$\frac{dh}{dt} = \frac{-15}{\pi h^2} \quad \text{(J1)}$$

$$\text{At } V = 3000\pi$$

$$3000 = \frac{\pi h^3}{9}$$

$$h = 30 \quad \text{(B1)}$$

$$\therefore \frac{dh}{dt} = \frac{-15}{\pi (30)^2} \quad \text{(K1) ✓}$$

$$= -\frac{1}{60\pi} \quad \text{or } -5.31 \times 10^{-3} \text{ cm s}^{-1} \quad \text{(J1)}$$

